

Vertex-minimal hyperbolic origami 2-torus

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Abstract

In this paper, we explore geometrizations of triangulated hyperbolic 2-tori Σ_2 in the hyperbolic 3-space $\mathbb{H}^3$. We show that for all $n \geq 10$, there exists a triangulation of Σ_2 with n vertices and an embedding $S : \Sigma_2 \hookrightarrow \mathbb{H}^3$ that restricts to isometries on the triangles. We call these embeddings *hyperbolic origami 2-tori*. Since the lower bound $n \geq 10$ is combinatorially sharp, this paper settles the question of minimum number of vertices required to obtain a hyperbolic embedding.

1 Introduction

1.1 Context and Problem Statement

A *hyperbolic 2-torus* Σ_2 is a quotient of $\mathbb{H}^2/\Gamma$ that is topologically an orientable closed surface of genus 2. Here, Γ is a discrete subgroup of $\text{Isom}(\mathbb{H}^2)$ isomorphic to the group with presentation $\langle a, b, c, d \mid [a, b][c, d] \rangle$, where $[a, b] = aba^{-1}b^{-1}$ denotes the commutator of the two elements, and the action of Γ on $\mathbb{H}^2$ is free and properly discontinuous. Every hyperbolic 2-torus Σ_2 inherits the structure of a Riemannian manifold of constant sectional curvature -1 from its universal cover. A *hyperbolic triangulation* $\mathcal{T}$ of Σ_2 is a triangulation of Σ_2 whose edges are all geodesics. A *hyperbolic origami 2-torus* is a map $S : (\Sigma_2, \mathcal{T}) \hookrightarrow \mathbb{H}^3$ that restricts to isometries on the triangles. Concretely, we think of a hyperbolic origami 2-torus S as gluing hyperbolic triangles together in $\mathbb{H}^3$, where the cone angles at each vertex are all 2π .

The study of graph embeddings and triangulations of closed surfaces can be dated back to the late 19th century [Hea90]. The minimum number of vertices n_0 to triangulate a closed surface of Euler characteristic χ is given by

$$n_0 \geq \left\lceil \frac{7 + \sqrt{49 - 24\chi}}{2} \right\rceil \quad (1)$$

Equation (1) is proved by Ringel [Rin55] for the non-orientable cases and later by Jungerman and Ringel [JR80] for the orientable cases. However, the bounds in Equation (1) is not tight for the 2-torus. In [Hum78], Huneke showed that the minimum number of vertices required to triangulate a 2-torus is 10, in contrast with a looser lower bound of 8 from Equation (1).

In 2008, Lutz [Lut08] enumerated all 865 isomorphism classes of 10-vertex triangulations of the 2-torus.

Geometers were interested in whether these triangulated surfaces admit piecewise-affine embeddings into $\mathbb{R}^3$ that are affine on the triangle pieces [BB87; Bre87; BB89; BZ95; HLZ06]. These embeddings are called *polyhedral embeddings*. A polyhedral embedding of a 10-vertex triangulation of the 2-torus was first discovered by Brehm [Bre81]. Later, Hougardy, Lutz, and Zelke [HLZ07] developed a computer algorithm to enumerate polyhedral embeddings of all 10-vertex triangulations of the 2-torus. However, a fundamental problem remains unsolved: if we take into account the hyperbolic structures on a 2-torus, these polyhedral embeddings fail to preserve them. First, it's impossible to map hyperbolic triangles isometrically to geodesic triangles in $\mathbb{R}^3$ due to Gauss-Bonnet theorem. Thus, it is more natural to consider embeddings of $(\Sigma_2, \mathcal{T})$ into $\mathbb{H}^3$, where the images of triangles in $\mathcal{T}$ are isometric geodesic triangles in the hyperbolic 3-space.

There has been significant progress on polyhedral embeddings of the flat torus. See [Zal00; BH08; Seg16; LT24; Tsu24]. A *flat polyhedral torus* is a piecewise-linear embedding of a triangulated torus T^2 into $\mathbb{R}^3$ that restricts to linear isometries on the triangles, where the triangles in T^2 come from a triangular tiling of the universal cover $\mathbb{R}^2$. A very recent work of Schwartz [Sch25] showed that a flat polyhedral torus of 7 vertices doesn't exist, but one can obtain one with 8 vertices with 2-fold symmetry. This settled the question of the minimum number of vertices required for a flat polyhedral torus. In this paper, we prove that the minimum number of vertices required for a hyperbolic origami 2-torus is 10.

1.2 The Candidate Model 2-Torus

We state the main result of this paper.

Theorem 1.1. *For all $n \geq 10$, there exists a triangulation $\mathcal{T}$ of Σ_2 with n vertices and a hyperbolic origami 2-torus $S : (\Sigma_2, \mathcal{T}) \hookrightarrow \mathbb{H}^3$.*

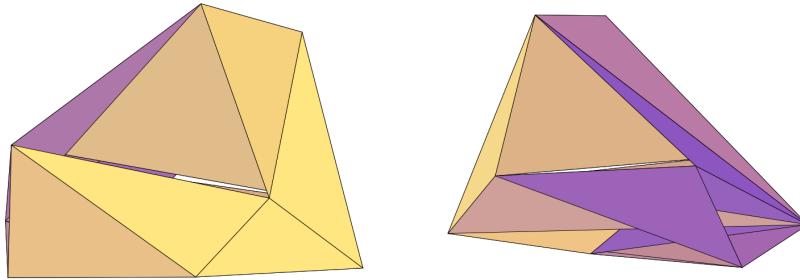


Figure 1: The candidate 2-torus $\hat{S}$ in the Beltrami–Klein model $\mathbb{H}^3$, shown from opposite sides of a “hole” passing through it. Triangles are shaded along a yellow–purple spectrum according to the y -coordinates of their centroids, ordered from highest to lowest.

The information of a hyperbolic origami 2-torus is captured entirely by the triangulation $\mathcal{T}$ and the image of the vertices of $\mathcal{T}$ under the map $S : (\Sigma_2, \mathcal{T}) \rightarrow \mathbb{H}^3$, as long as the cone angles of the vertices of $\mathcal{T}$ are all 2π . We will show the existence of a 10-vertex triangulation

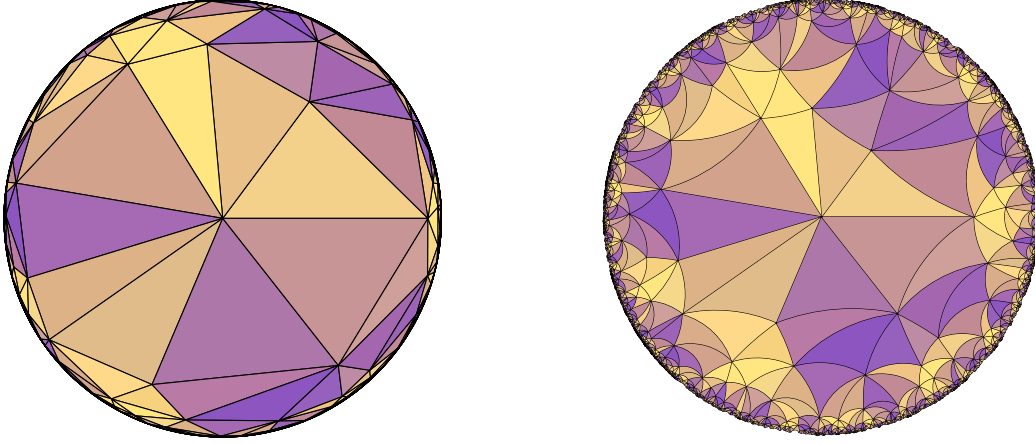


Figure 2: Left: The induced triangle tiling from $\mathcal{T}$ in the Beltrami–Klein model of $\mathbb{H}^2$. Right: The tiling in the Poincaré disk model. The colors of the triangles are derived from Figure 1.

$\mathcal{T}$ and a hyperbolic origami 2-torus S . To get a hyperbolic origami 2-torus with arbitrary number of vertices $n \geq 10$, we can add vertices to the interior of the triangles in $\mathcal{T}$ to obtain triangulations $\mathcal{T}'$ with more vertices. The map $S : \Sigma_2 \rightarrow \mathbb{H}^3$ remains a hyperbolic origami 2-torus with respect to the subdivided triangulations $\mathcal{T}'$.

Consider the following 10-vertex triangulation $\mathcal{T} = (V, E, F)$ with vertices in V , edges in E , and oriented triangles in F . We index V by $\{0, \dots, 9\}$. We list the degree sequence of the 10 vertices in V :

$$(d_0, d_1, d_2, d_3, d_4, d_5, d_6, d_7, d_8, d_9) = (9, 8, 8, 8, 8, 7, 7, 6, 6, 5).$$

We also list out the indices of the 24 oriented triangles in F in the form of 3-tuples. A tuple (i, j, k) indicates an oriented face joined by vertices i, j, k .

$$\begin{array}{cccccc} (0, 1, 7) & (0, 2, 1) & (0, 3, 6) & (0, 4, 9) & (0, 5, 3) & (0, 6, 4) \\ (0, 7, 8) & (0, 8, 5) & (0, 9, 2) & (1, 2, 4) & (1, 3, 7) & (1, 4, 6) \\ (1, 5, 8) & (1, 6, 5) & (1, 8, 3) & (2, 3, 8) & (2, 6, 3) & (2, 7, 4) \\ (2, 8, 7) & (2, 9, 6) & (3, 4, 7) & (3, 5, 4) & (4, 5, 9) & (5, 6, 9) \end{array} \quad (2)$$

The proof of Theorem 1.1 consists of two steps. First, we find an embedded hyperbolic origami 2-torus $\hat{S}$ that is “robustly embedded” and “almost flat.” Next, we show that we can get a flat one by perturbing the coordinates of $\hat{S}$. Figure 1 shows the hyperbolic polyhedral 2-torus $\hat{S}$ embedded in the Beltrami–Klein model of the hyperbolic 3-space, where the geodesics are straight lines and geodesic triangles are triangles in the Euclidean sense. In Figure 2 we attempt to draw the induced tiling of the universal cover $\mathbb{H}^2$ by the hyperbolic triangulation on $\mathcal{T}$ on Σ_2 by isometrically “lifting” the triangles in $\hat{S}$ to $\mathbb{H}^2$ and gluing them together. Since $\hat{S}$ is not an origami 2-torus, we can only draw the tiling in the universal cover up to some error (about 10^{-28}), but the error is too small to be discernable by human eyes so that we can still make sense of the approximated tiling. See Figure 3, 4, 5 for the intersection of $\hat{S}$ with the xy -plane, xz -plane, and yz -plane.

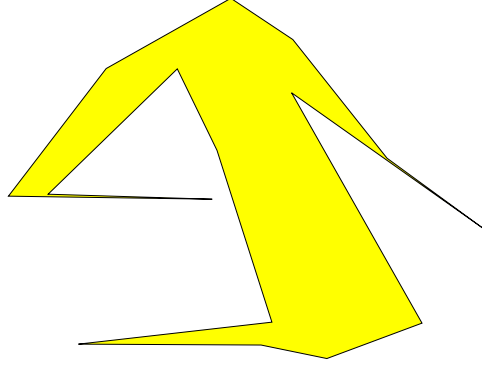


Figure 3: The intersection of $\hat{S}$ with the xy -plane, which is the boundary of a simply-connected component colored in yellow.

Here we give the exact coordinates of the vertices $\hat{X}_0, \dots, \hat{X}_9$ of our candidate 2-torus $\hat{S}$ in the Beltrami–Klein model:

0.7315	0.0202	0.28688022781563440615364787558404	
−0.316	0.5792	−0.2252919753182150895621576631383	
0.3426	−0.592	−0.22851917827575874828458055465199	
−0.4323	−0.592	−0.23272863894943839798113773793091	
−0.7303	0.04	−0.22959077803009316431117678104662	
0.1464	0.6149	0.13588682780065943976868637469759	(3)
−0.5154	0.0395	0.46102777383206591809202059407883	
0.6649	−0.1156	−0.22651115997910956793325851442687	
0.152	0.2539	−0.23985732806740791506457015735734	
−0.03	0.0606	0.64396456614136038886542316026992	

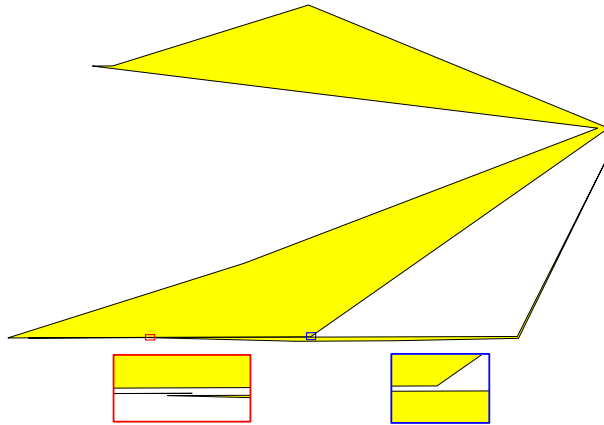


Figure 4: The intersection of $\hat{S}$ with the plane $y = 0$, which is the boundary of two simply-connected components colored in yellow. The boxes contain the manifold view of their respective regions in the original slice.

We explain how we chose the triangulation $\mathcal{T}$ and obtained these coordinates. We start with all triangulations $\mathcal{T}_n = (V_n, E_n, F_n)$ and polyhedral embeddings $S_n : (\Sigma_2, \mathcal{T}_n) \hookrightarrow \mathbb{R}^3$ from [HLZ07] for all 865 triangulations $n = 1, \dots, 865$, where S_n maps vertices of $\mathcal{T}_n$ to the integer lattice $\mathbb{Z}^3 \cap [0, 4]^3$. First, we apply a translation and scale the vectors so that the center-of-mass of the vertices is at the origin, and that $\max_{X \in V_n} \|X\| = \frac{1}{2}$. This ensures that $S_n(\Sigma_2) \subset B_1(0)$, so we can treat S_n as an embedding into the Beltrami–Klein model of $\mathbb{H}^3$. Then, for each n , we compute the hyperbolic angles of all triangles in S_n and all the cone angles. We found that S_{265} has the minimal cone angle difference to 2π among all other embeddings. After that, we apply a hill-climbing algorithm by randomly perturbing the coordinates of S_{265} to minimize the objective function

$$\mathcal{L}(S) = \max_{i=0, \dots, 9} |\theta_i(S) - 2\pi|$$

where $\theta_i(S)$ is the cone angle of S at vertex i . This gives us a surface S'_{265} with $\mathcal{L}(S'_{265}) \leq 10^{-14}$. Let $z = (z_0, \dots, z_9)$ denote the vector of z -coordinates of S . We implement a high-precision version of Newton's method on the multivariable function

$$\Theta(z) = (\theta_0(z) - 2\pi, \theta_1(z) - 2\pi, \dots, \theta_9(z) - 2\pi). \quad (4)$$

We obtain the coordinates of $\hat{S}$ by truncating the z -coordinates at 10^{-32} .

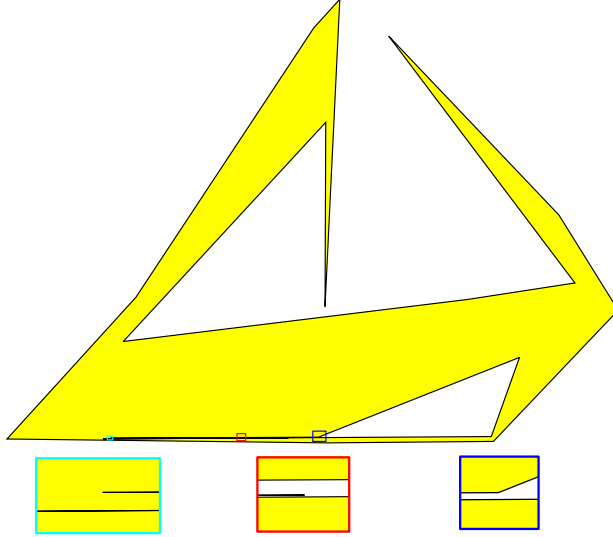


Figure 5: The intersection of $\hat{S}$ with the plane $x = 0$, which is the boundary of a region that is not simply-connected. Again, the boxes contain the manifold view of their respective regions in the original slice.

This paper is organized as follows: In §2, we discuss the properties of the Beltrami–Klein model and explain how we incorporate floating-point arithmetic into the proofs. In §3, we show that the cone angles of $\hat{S}$ differ from 2π by at most 10^{-28} , and that $\hat{S}$ stays embedded after perturbing its z -coordinates by at most 10^{-7} . In §4, we analyze the Jacobian matrix of $\Theta(z)$ to show that there exists a flat embedded hyperbolic origami 2-torus S near $\hat{S}$.

1.3 Acknowledgements

This project is supported by an N.S.F. Research Grant DMS-2505281. I would like to thank my advisor, Richard Schwartz, for introducing this problem to me and kindly explaining to me his proof of the minimal-vertex paper torus problem. Much of the methodology used in this paper is inspired by [Sch25]. I would like to thank Alba Málaga, Fabian Lander, Stepan Paul, and Steve Trettel for helpful discussions on this topic. I would like to thank Frank Lutz for his manifolds webpage and Stefan Hougardy, Frank H. Lutz, and Mariano Zelke for keeping a database of piecewise-affine embedded 2-tori with vertex-minimal triangulations, which helped me find prototypes for the candidate 2-torus in this paper.

2 Background

2.1 The Beltrami–Klein Model

Denote by $\mathbb{H}^n$ the hyperbolic n -space – the simply connected Riemannian n -manifold of constant sectional curvature -1 . We mainly work with $\mathbb{H}^3$ in this paper. Here we provide the explicit construction known as the *Beltrami–Klein model* of $\mathbb{H}^3$. Consider the open unit Euclidean ball $B_1(0) \subset \mathbb{R}^3$. For each $X \in B_1(0)$, we write $a_X = 1 - \|X\|^2$. The Riemannian manifold $\mathbb{H}^3$ is defined as the manifold $B_1(0)$ along with the Riemannian metric g given by

$$g|_X = \left(\frac{(dx^1)^2 + (dx^2)^2 + (dx^3)^2}{a_X} + \frac{(x^1 dx^1 + x^2 dx^2 + x^3 dx^3)^2}{a_X^2} \right) \quad (5)$$

Let V, W be two arbitrary vectors in $\mathbb{R}^3$, which we can identify with $T_X \mathbb{H}^3$. Then, $g|_X$ can be expressed as

$$g|_X(V, W) = \frac{\langle V, W \rangle}{a_X} + \frac{\langle X, V \rangle \langle X, W \rangle}{a_X^2} = \frac{a_X \langle V, W \rangle + \langle X, V \rangle \langle X, W \rangle}{a_X^2}. \quad (6)$$

where $\langle \cdot, \cdot \rangle$ denotes the Euclidean dot product. We will write $\langle V, W \rangle_X = g|_X(V, W)$ and $\|V\|_X = \sqrt{g|_X(V, V)}$ the norm of $V \in T_X \mathbb{H}^n$.

Since geodesics in $\mathbb{H}^3$ are straight lines, a totally geodesic triangle in $\mathbb{H}^3$ with vertices (X, Y, Z) is the same as the Euclidean triangle in the unit ball. We may compute the hyperbolic angle θ at X as follows: Let $V = Y - X$ and $W = Z - X$. We may identify V and W with velocity vectors in $T_X \mathbb{H}^3$ of the geodesic from X to Y and X to Z respectively. Consider the following function:

$$A(X, Y, Z) = \frac{\langle V, W \rangle_X^2}{\langle V, V \rangle_X \cdot \langle W, W \rangle_X} = \frac{(a_X \langle V, W \rangle + \langle X, V \rangle \langle X, W \rangle)^2}{(a_X \langle V, V \rangle + \langle X, V \rangle^2)(a_X \langle W, W \rangle + \langle X, W \rangle^2)}. \quad (7)$$

Denote by $\sigma = \text{sgn}(\langle V, W \rangle_X)$ the sign of $\langle V, W \rangle_X$. It follows that

$$\theta = \arccos \left(\frac{\langle V, W \rangle_X}{\|V\|_X \|W\|_X} \right) = \arccos \left(\sigma \sqrt{A(X, Y, Z)} \right); \quad \cos \theta = \sigma \sqrt{A(X, Y, Z)}. \quad (8)$$

2.2 Remark on Floating-Point Arithmetic

The proofs in this paper make use of computer floating-point arithmetic. To perform floating-point arithmetic up to high precision (10^{-400} for our purposes), we use the python package `mpmath`, which stores floating-point numbers of any designated precision and has built-in methods of basic arithmetic operations. When we claim that $\frac{a}{b} < \frac{c}{d}$ for floating-point numbers a, b, c, d , we implicitly carry out the computation of $ad - bc$ in `mpmath` and check that the result is negative. We sometimes find equivalent fractions $\frac{a'}{b'} = \frac{a}{b}$ and $\frac{c'}{d'} = \frac{c}{d}$ with $a', b', c', d' \in \mathbb{Z}$ to simplify the computation into integer arithmetic.

For finding the bounds on the value of e^a where a is a floating-point number, see §5.1.

3 The Candidate 2-Torus

In this section, we show that the cone angles of our candidate 2-torus $\hat{S}$ are very close to 2π . We then show that if we apply perturbation to the z -coordinates of $\hat{S}$, the resulting 2-torus stays embedded as long as the perturbation is small enough.

Let $\mathcal{T}$ be a triangulation of the 2-torus Σ_2 . A *hyperbolic polyhedral 2-torus* is an embedding $S : (\Sigma_2, \mathcal{T}) \hookrightarrow \mathbb{H}^3$ such that the image of the triangles in $\mathcal{T}$ under S are geodesic triangles. A hyperbolic origami 2-torus is then a hyperbolic polyhedral 2-torus where the cone angle of each vertex is 2π .

3.1 Robust Flatness

We start by defining flatness of a hyperbolic polyhedral 2-torus.

Definition 3.1. We say that a hyperbolic polyhedral 2-torus S is ε -flat if $\max_i |\theta_i - 2\pi| < \varepsilon$, where $i \in \{0, \dots, 9\}$ and θ_i is the cone angle of vertex i .

Proposition 3.2. $\hat{S}$ is 10^{-28} -flat.

Proof. Let d_i be the degree of vertex i , and let $n_0, \dots, n_{d_i-1}$ be the indices of the neighbors of i , ordered so that $(i, n_j, n_{j+1}) \in F$ for $j = 0, \dots, d_i - 1$ (here the subscripts j and $j + 1$ are taken modulo d_i). We first find integer vectors $Y_{i,n_j} \in \mathbb{R}^2$ for all $i = 0, \dots, 9$ and $j = 0, \dots, d_i - 1$ so that the difference $\delta_{i,j} = |\alpha_{i,j} - \beta_{i,j}|$ is minimized, where

$$\alpha_{i,j} = A(\hat{X}_i, \hat{X}_{n_j}, \hat{X}_{n_{j+1}}); \quad \beta_{i,j} = \frac{\langle Y_{i,n_j}, Y_{i,n_{j+1}} \rangle^2}{\langle Y_{i,n_j}, Y_{i,n_j} \rangle \langle Y_{i,n_{j+1}}, Y_{i,n_{j+1}} \rangle}$$

Here, A is the map from Equation (7). The coordinates of Y_{i,n_j} are given in §5.2. We obtained these coordinates by lifting the link of each vertex $\hat{X}_i$ isometrically into the Poincaré model of the universal cover $\mathbb{H}^2$, with $\hat{X}_i \mapsto 0$ and $\hat{X}_{n_0}$ mapping to the positive real axis. Then, we scale up each coordinate by 10^{32} to make them all integer-valued. We computed using floating-point arithmetic that $\max_{i,j} \delta_{i,j} \leq 2.93 \times 10^{-32}$. We also computed that $\alpha_{i,j} \in [0.000052, 0.918]$. It follows that $\alpha_{i,j}, \beta_{i,j} \in [0.00005, 0.92]$ for all i, j . In particular, $\langle X_{n_j} - X_i, X_{n_{j+1}} - X_i \rangle_{X_i}$ and $\langle Y_{i,n_j}, Y_{i,n_{j+1}} \rangle$ always share the same sign.

Next, we explain how we obtain the cone angle bounds from $\alpha_{i,j}$ and $\beta_{i,j}$. Consider two functions ϕ_+ and ϕ_- given by $\phi_+(x) = \arccos(\sqrt{x})$ and $\phi_-(x) = \arccos(-\sqrt{x})$. Denote by $\theta_{i,j}$ the hyperbolic angle at $\hat{X}_i$ of the triangle $(\hat{X}_i, X_{n_j}, \hat{X}_{n_{j+1}})$, and denote by $\tau_{i,j}$ the Euclidean angle between the two vectors Y_{i,n_j} and $Y_{i,n_{j+1}}$. Since $\langle X_{n_j} - X_i, X_{n_{j+1}} - X_i \rangle_{X_i}$ and $\langle Y_{i,n_j}, Y_{i,n_{j+1}} \rangle$ always share the same sign, we know that

- $\theta_{i,j} = \phi_+(\alpha_{i,j})$ and $\tau_{i,j} = \phi_+(\beta_{i,j})$ if $\langle X_{n_j} - X_i, X_{n_{j+1}} - X_i \rangle_{X_i} > 0$;
- $\theta_{i,j} = \phi_-(\alpha_{i,j})$ and $\tau_{i,j} = \phi_-(\beta_{i,j})$ if $\langle X_{n_j} - X_i, X_{n_{j+1}} - X_i \rangle_{X_i} < 0$.

Next, observe that

$$\frac{d\phi_+}{dx} = \frac{-1}{2\sqrt{x(1-x)}}; \quad \frac{d\phi_-}{dx} = \frac{1}{2\sqrt{x(1-x)}}$$

One can compute directly that $|\frac{d\phi_+}{dx}|, |\frac{d\phi_-}{dx}| \leq 71$ on $[0.00005, 0.92]$, so both ϕ_+ and ϕ_- are 71-Lipschitz on $[0.00005, 0.92]$. Finally, we have

$$\begin{aligned} |\theta_i - 2\pi| &= \left| \sum_{j=0}^{d_i-1} \theta_{i,j} - \sum_{j=0}^{d_i-1} \tau_{i,j} \right| \leq \sum_{j=0}^{d_i-1} |\theta_{i,j} - \tau_{i,j}| \leq \max_i d_i \cdot \max_{i,j} |\theta_{i,j} - \tau_{i,j}| \\ &\leq \max_i d_i \cdot 71 \cdot \max_{i,j} \delta_{i,j} \leq 71 \times 9 \times 2.93 \times 10^{-32} < 10^{-28}. \end{aligned}$$

Taking $\varepsilon = 10^{-28}$ completes the proof. $\square$

3.2 Robust Embeddedness

The next step is to show that if we slightly perturb the z -coordinates of $\hat{S}$, the resulting 2-torus is still embedded. The key observation (and the main reason we use the Beltrami–Klein model of the hyperbolic plane) is that the underlying set of the hyperbolic geodesic triangle with vertices $V_1, V_2, V_3 \in \mathbb{H}^3$ is precisely the Euclidean triangle with the same vertices. Therefore, in this section we “forget” the hyperbolic structure and instead treat $\hat{S}$ as a map $\hat{S} : (\Sigma_2, \mathcal{T}) \rightarrow B_1(0)$ whose image is the unit Euclidean ball. We begin with the following definition, which is not “natural” in the hyperbolic setting but is more appropriate for Euclidean geometry.

Definition 3.3. A map $S : (\Sigma_2, \mathcal{T}) \rightarrow B_1(0)$ is λ -robustly embedded if for all S' obtained from moving each of the z -coordinate of the vertices of S by at most λ is also embedded.

Remark 3.4. Readers may have noticed that Definition 3.3 is not invariant under hyperbolic isometries. That is, suppose $S : (\Sigma_2, \mathcal{T}) \hookrightarrow \mathbb{H}^3$ is λ -robustly embedded and $M \in \text{Isom}(\mathbb{H}^3)$, then there is no reason whatsoever to expect that $M \circ S$ must also be λ -robustly embedded. One should keep in mind that throughout this section, we will “pretend to” work with Euclidean geometry.

To show embeddedness, it suffices to consider pairs of triangles with no common vertices and pairs that share one vertex in common. Below is a lemma from [Sch25]. We present a self-contained proof here.

Lemma 3.5 ([Sch25, Lemma 3.2]). *Given a map $S : (\Sigma_2, \mathcal{T}) \rightarrow B_1(0)$ sending triangles in $\mathcal{T}$ to Euclidean triangles in $B_1(0)$. If all pairs of triangles of S sharing no vertices in common are disjoint and all pairs of triangles sharing one vertex only intersect at the shared vertex, then S is embedded.*

Proof. Consider the pairs of triangles (i, j, k) and (j, i, l) that share two vertices. Denote by e_{ab} the edge connecting X_a to X_b , where $a, b \in \{i, j, k, l\}$. We first recall that every edge is contained in two triangles, so the edge e_{ik} is contained in another triangle (i, k, m) , where $m \neq j$. If the interior of e_{ik} intersects (j, i, l) , then (j, i, l) and (i, k, m) intersect at a point other than X_i . Notice that $l \neq m$, for otherwise the degree of vertex i would equal 3. This implies (j, i, l) and (i, k, m) only share one vertex, but that contradicts our assumption in the lemma. The same argument shows the interior of e_{jk} doesn't intersect (j, i, l) , and the interior of e_{il} and e_{jl} doesn't intersect (i, j, k) , so these two triangles only intersect at e_{ij} , which shows that S is embedded. $\square$

We first deal with the pairs of disjoint triangles. We say that two disjoint triangles T_1 and T_2 in $\mathbb{R}^3$ are δ -separated if they remain disjoint after simultaneously perturbing the z -coordinates of the vertices by at most δ . Here is a sufficient hyperplane separation test to show that T_1 and T_2 are δ -separated.

Lemma 3.6. *Two disjoint triangles T_1 and T_2 are δ -separated if there exists a vector $\hat{N}$ with $\|\hat{N}\|_\infty < C$ for some $C \in \mathbb{R}$, such that*

$$\min_{X \in T_1} \langle X, \hat{N} \rangle - \max_{Y \in T_2} \langle Y, \hat{N} \rangle > 2\delta C. \quad (9)$$

In particular, if T_1 has vertices V_1, V_2, V_3 and T_2 has vertices W_1, W_2, W_3 , then it suffices to check that

$$\min_{X \in \{V_1, V_2, V_3\}} \langle X, \hat{N} \rangle - \max_{Y \in \{W_1, W_2, W_3\}} \langle Y, \hat{N} \rangle > 2\delta C. \quad (10)$$

Proof. We know that every $X \in T$ can be expressed as $X = t_1 V_1 + t_2 V_2 + t_3 V_3$ where $t_i \geq 0$ and $t_1 + t_2 + t_3 = 1$. Linearity of the dot product then implies the minimum of $\langle X, \hat{N} \rangle$ must be achieved at one of the vertices of T , so Equation (10) implies (9). Now, if T'_1 and T'_2 are obtained from perturbing the z -coordinates of the vertices of T_1 and T_2 by at most δ , then the dot product $\langle V_i, \hat{N} \rangle$ is perturbed by at most δC . It follows that

$$\min_{X' \in T'_1} \langle X', \hat{N} \rangle - \max_{Y' \in T'_2} \langle Y', \hat{N} \rangle \geq \min_{X \in T_1} \langle X, \hat{N} \rangle - \max_{Y \in T_2} \langle Y, \hat{N} \rangle - 2\delta C > 0.$$

This implies $T'_1 \cap T'_2 = \emptyset$, which completes the proof. $\square$

Next, we deal with triangles that share one of their vertices. Suppose (U, V_1, V_2, W_1, W_2) are five points in $\mathbb{R}^3$, and T_1, T_2 are two triangles with vertices (U, V_1, V_2) and (U, W_1, W_2) respectively. We say that T_1 and T_2 are δ -separated if for all T'_1 and T'_2 with vertices (U', V'_1, V'_2) and (U', W'_1, W'_2) such that $(U', V'_1, V'_2, W'_1, W'_2)$ are obtained from perturbing the z -coordinates of (U, V_1, V_2, W_1, W_2) by at most δ , $T'_1 \cap T'_2 = \{U'\}$.

Lemma 3.7. T_1 and T_2 are δ -separated if there exists a vector $\hat{N}$ with $\|\hat{N}\|_\infty < C$ for some $C \in \mathbb{R}$, such that

$$\min_{X \in \{V_1, V_2\}} \langle X, \hat{N} \rangle - \langle U, \hat{N} \rangle > 2\delta C \text{ and } \langle U, \hat{N} \rangle - \min_{Y \in \{W_1, W_2\}} \langle Y, \hat{N} \rangle > 2\delta C. \quad (11)$$

Proof. Suppose T'_1 and T'_2 are obtained from perturbing (U, V_1, V_2, W_1, W_2) by at most δ . The dot products with $\hat{N}$ is perturbed by at most δC . It follows that

$$\min_{X' \in \{V'_1, V'_2\}} \langle X', \hat{N} \rangle - \langle U', \hat{N} \rangle > 0 \text{ and } \langle U', \hat{N} \rangle - \min_{Y' \in \{W'_1, W'_2\}} \langle Y', \hat{N} \rangle > 0. \quad (12)$$

To see this implies $T'_1 \cap T'_2 = \{U'\}$, suppose there exists $X' \in T'_1 \cap T'_2$. We may express $X' = t_0U + t_1V'_1 + t_2V'_2 = s_0U + s_1W'_1 + s_2W'_2$ where $t_0 + t_1 + t_2 = s_0 + s_1 + s_2 = 1$ and all coefficients $t_i, s_j \geq 0$. It follows that

$$\langle X', \hat{N} \rangle - \langle U', \hat{N} \rangle = t_1 \langle V'_1 - U', \hat{N} \rangle + t_2 \langle V'_2 - U', \hat{N} \rangle$$

and

$$\langle U', \hat{N} \rangle - \langle X', \hat{N} \rangle = s_1 \langle U', -W'_1, \hat{N} \rangle + s_2 \langle U' - W'_2, \hat{N} \rangle.$$

This implies

$$t_1 \langle V'_1 - U', \hat{N} \rangle + t_2 \langle V'_2 - U', \hat{N} \rangle + s_1 \langle U', -W'_1, \hat{N} \rangle + s_2 \langle U' - W'_2, \hat{N} \rangle = 0,$$

so Equation (12) implies $t_1 = t_2 = s_1 = s_2 = 0$, which gives us $X' = U'$ as desired. $\square$

We are now ready to prove that $\hat{S}$ is robustly embedded.

Proposition 3.8. $\hat{S}$ is 10^{-7} -robustly embedded.

Proof. We dilate the surface $\hat{S}$ by a factor of 10^{32} so that every vertex has integer coordinates. Call the dilated surface $\hat{S}'$. Lemma 3.6 and 3.7 tells us that it suffices to find normal vectors $\hat{N}$ for each pair of triangles in $\hat{S}'$.

Among the 276 pairs of triangles from F , there are 82 pairs that share no vertices in common, and 158 of which share one vertex in common. We provide an algorithm to generate $\hat{N}$ for each of these pairs where the conditions in Lemma 3.6 and 3.7 holds. Consider a map $\rho : \mathbb{Z}_{>0} \rightarrow \mathbb{Z}^3$ as follows:

$$\rho(n) = \left(\lfloor 10^5 \cdot L(n\sqrt{2}) \rfloor, \lfloor 10^5 \cdot L(n\sqrt{3}) \rfloor, \lfloor 10^5 \cdot L(n\sqrt{5}) \rfloor \right); \quad L(x) = 2(x - \lfloor x \rfloor) - 1$$

It is clear that $\|\rho(n)\|_\infty < 10^5$ for all n . We found indices $n < 2000$ such that the conditions in Lemma 3.6 hold for $\delta = 10^{25}$ by taking $\hat{N} = \pm \rho(n)$ for all but the following two pairs of disjoint triangles, where we manually found the candidates for $\hat{N}$:

$$\begin{aligned} \{(2, 7, 4), (1, 8, 3)\}, \quad \hat{N} &= (-35, -74, 12106); \\ \{(0, 2, 1), (3, 4, 7)\}, \quad \hat{N} &= (-60, -96, 22534). \end{aligned}$$

We also computed $\rho(n)$ for the triangle pairs that share one vertex. We found indices $n < 10^5$ such that the conditions in Lemma 3.7 hold for $\delta = 10^{25}$ by taking $\hat{N} = \pm\rho(n)$ for all but the following nine pairs, where we manually found the candidates for $\hat{N}$:

$$\begin{aligned}
&\{(0, 2, 1), (1, 3, 7)\}, \quad \hat{N} = (-40, -67, 14035); \\
&\{(0, 2, 1), (2, 7, 4)\}, \quad \hat{N} = (73, 97, -22643); \\
&\{(1, 2, 4), (1, 3, 7)\}, \quad \hat{N} = (-54, -85, 14773); \\
&\{(1, 2, 4), (1, 8, 3)\}, \quad \hat{N} = (-120, -151, 27015); \\
&\{(1, 2, 4), (3, 4, 7)\}, \quad \hat{N} = (45, 69, -14773); \\
&\{(1, 3, 7), (2, 3, 8)\}, \quad \hat{N} = (-184, -155, -15412); \\
&\{(1, 3, 7), (2, 7, 4)\}, \quad \hat{N} = (-36, -73, 12086); \\
&\{(1, 8, 3), (3, 4, 7)\}, \quad \hat{N} = (-35, -74, 12107); \\
&\{(2, 3, 8), (3, 4, 7)\}, \quad \hat{N} = (-417, 566, 51293).
\end{aligned}$$

Lemma 3.5 then implies the 10^{32} -diluted surface $\hat{S}'$ is 10^{25} -robustly embedded, so the original surface $\hat{S}$ is 10^7 -robustly embedded, as desired. $\square$

4 Finding Origami 2-Tori

Our method of finding the flat 2-tori is analogous to Schwartz's method in [Sch25]. The idea is to use the inverse function theorem to show that there exists a neighborhood around $\hat{z}$ whose image under the map Θ in Equation (4) hits the origin.

Definition 4.1. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a smooth map. We say that f is λ -*expansive* on a subset $A \subset \mathbb{R}^n$ if for all unit vectors $V \in \mathbb{R}^n$ and $p, q \in A$, we have

- $\|df|_p(V)\| > 2\lambda$;
- The angle between $df|_p(V)$ and $df|_q(V)$ is less than $\frac{\pi}{3}$.

The following lemma is proved by Schwartz in [Sch25]. Here we provide an alternative self-contained proof.

Lemma 4.2 ([Sch25, Lemma 3.3]). *Let $B_r(p) \subset \mathbb{R}^n$ denote the open ball of radius r centered at p . If $G : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is λ -expansive on $B_r(p)$ for some $\lambda > 0$, then $B_{\lambda r}(G(p)) \subset G(B_r(p))$.*

Proof. After normalizing by translation and scaling we may assume $f(p) = p = 0$, $\lambda = r = 1$. Since $\|df|_q(V)\| > 0$ for all $q \in B_1(0)$ and unit vectors V , $df|_q$ is nondegenerate, so by the inverse function theorem, f is an open map on $B_1(0)$. It follows that $\mathbb{R}^n \setminus f(B_1(0))$ is closed. Consider the set $A = \overline{B_s(0)} \setminus f(B_1(0))$, where $s = 2 \sup_{x \in B_1(0)} \|f(x)\|$. We know that A is compact, and $A \cup f(B_1(0)) = \overline{B_s(0)}$. The statement is true if $\|x\| \geq 1$ for all $x \in A$.

Since A is compact, $\inf_{x \in A} \|x\|$ is achieved by some $\hat{x} \in A$. Assume for contradiction that $\|\hat{x}\| = \alpha$ for some $\alpha < 1$. Then, for all $n \in \mathbb{N}$, there exists $x_n \in B_1(0)$

such that $f(x_n) \in B_{1/n}(\hat{x})$. It follows that $\lim_{n \rightarrow \infty} f(x_n) = \hat{x}$. On the other hand, let $\beta[0, 1] \rightarrow \mathbb{R}^n$ be the unit-speed parametrization of the line segment from 0 to x_n , and let $\gamma = f \circ \beta$. Since $\|df|_{\gamma(t)}(\beta'(t))\| > 2$, we know that γ has length at least $2\|x_n\|$. Also, since $\langle df|_{\gamma(t)}(\beta'(t)), df|_0(\beta'(t)) \rangle < \frac{\pi}{3}$, the projection of $\gamma(t)$ onto the line connecting 0 to $f(x_n)$ is a curve that moves at a speed greater than 1. It follows that $\|f(x_n)\| > \|x_n\|$, so $\|x_n\| < \frac{1+\alpha}{2}$ for n sufficiently large. This implies the sequence $\{x_n\}$ lives in a compact subset of $B_1(0)$, so it converges to some $y \in B_1(0)$ on a subsequence, but then $f(y) = \hat{x}$, contradicting $\hat{x}_n \in A$. $\square$

Denote by $\mathcal{B}$ the set of embeddings S whose z -coordinates lie in $B_{10^{-18}}(\hat{z})$, where $\hat{z}$ is the vector of z -coordinates of the almost-flat 2-torus $\hat{S}$ from §3.

Proposition 4.3. *The map $\Theta : \mathbb{R}^{10} \rightarrow \mathbb{R}^{10}$ is $\frac{1}{2}$ -expansive on $B_{10^{-18}}(\hat{z})$.*

Proof of Theorem 1.1 assuming Proposition 4.3. From Proposition 3.2 we know that $\hat{S}$ is 10^{-28} -flat, so $\|\Theta(\hat{z})\| < 10^{-27}$, or $0 \in B_{10^{-27}}(\Theta(\hat{z}))$. Then, Proposition 4.3 implies the existence of a hyperbolic origami 2-torus S with coordinates $z \in B_{5 \times 10^{-19}}(\hat{z})$ such that $\Theta(z) = 0$. In particular, Proposition 3.8 implies S is embedded. This yields the hyperbolic origami 2-torus as desired. $\square$

The rest of this section will be devoted to proving Proposition 4.3.

4.1 The Jacobian

Our first goal is to analytically compute the Jacobian matrix of the map Θ . Recall from Equation (4) that Θ is given by

$$\Theta_i(z) = \sum_{(i,j,k) \in F} \theta_{ijk}(z) - 2\pi,$$

where $\theta_{ijk}(z)$ denote the angle of the triangle (i, j, k) at vertex i .

We start by computing the first-order partials of θ_{ijk} . Given $X_i = (x_i, y_i, z_i)$, denote by ∂_i the operator $\frac{\partial}{\partial z_i}$. Consider a triangle $(i, j, k) \in F$. We again write $V = X_j - X_i$, $W = X_k - X_i$, and $a_{X_i} = 1 - \|X_i\|^2$. We would like to compute the action of $\partial_i, \partial_j, \partial_k$ on $\langle V, W \rangle_{X_i}$, $\|V\|_{X_i}$, and $\|W\|_{X_i}$. By symmetry of the Riemannian metric, it suffices to compute ∂_i and ∂_j . We first observe that

$$\begin{aligned} \partial_i \langle X_i, X_i \rangle &= 2z_i & \partial_i \langle V, V \rangle &= 2z_i - 2z_j & \partial_i \langle W, W \rangle &= 2z_i - 2z_k \\ \partial_i \langle X_i, V \rangle &= z_j - 2z_i & \partial_i \langle X_i, W \rangle &= z_k - 2z_i & \partial_i \langle V, W \rangle &= 2z_i - z_j - z_k \\ \partial_j \langle V, X_i \rangle &= z_i & \partial_j \langle V, W \rangle &= z_k - z_i & \partial_j \langle V, V \rangle &= 2z_j - 2z_i \end{aligned}$$

It follows that

$$\begin{aligned} \partial_i \langle V, W \rangle_{X_i} &= \partial_i \left(\frac{\langle V, W \rangle}{a_{X_i}} + \frac{\langle X_i, V \rangle \langle X_i, W \rangle}{a_{X_i}^2} \right) \\ &= \frac{2z_i - z_j - z_k}{a_{X_i}} + \frac{2z_i \langle V, W \rangle + (z_j - 2z_i) \langle X_i, W \rangle + (z_k - 2z_i) \langle X_i, V \rangle}{a_{X_i}^2} + \frac{4z_i \langle X_i, V \rangle \langle X_i, W \rangle}{a_{X_i}^3} \end{aligned} \quad (13)$$

and

$$\partial_j \langle V, W \rangle_{X_i} = \partial_j \left(\frac{\langle V, W \rangle}{a_{X_i}} + \frac{\langle X_i, V \rangle \langle X_i, W \rangle}{a_{X_i}^2} \right) = \frac{z_k - z_i}{a_{X_i}} + \frac{z_i \langle X_i, W \rangle}{a_{X_i}^2}. \quad (14)$$

We also have

$$\begin{aligned} \partial_i \|V\|_{X_i} &= \partial_i \sqrt{\langle V, V \rangle_{X_i}} = \frac{1}{2\|V\|_{X_i}} \partial_i \langle V, V \rangle_{X_i} \\ &= \frac{1}{\|V\|_{X_i}} \left(\frac{z_i - z_j}{a_{X_i}} + \frac{z_i \langle V, V \rangle + (z_j - 2z_i) \langle X_i, V \rangle}{a_{X_i}^2} + \frac{2z_i \langle X_i, V \rangle^2}{a_{X_i}^3} \right) \end{aligned} \quad (15)$$

and

$$\partial_j \|V\|_{X_i} = \frac{1}{2\|V\|_{X_i}} \partial_j \langle V, V \rangle_{X_i} = \frac{1}{\|V\|_{X_i}} \left(\frac{z_j - z_i}{a_{X_i}} + \frac{z_i \langle X_i, V \rangle}{a_{X_i}^2} \right). \quad (16)$$

Therefore, for all $l \in \{i, j, k\}$, we have

$$\begin{aligned} \partial_l \theta_{ijk} &= \partial_l \arccos \left(\frac{\langle V, W \rangle_{X_i}}{\|V\|_{X_i} \|W\|_{X_i}} \right) = \frac{-\|V\|_{X_i} \|W\|_{X_i}}{\sqrt{\langle V, V \rangle_{X_i} \langle W, W \rangle_{X_i} - \langle V, W \rangle_{X_i}^2}} \cdot \partial_l \left(\frac{\langle V, W \rangle_{X_i}}{\|V\|_{X_i} \|W\|_{X_i}} \right) \\ &= \frac{\partial_l (\|V\|_{X_i} \|W\|_{X_i}) \langle V, W \rangle_{X_i}}{\|V\|_{X_i} \|W\|_{X_i} \cdot \sqrt{\langle V, V \rangle_{X_i} \langle W, W \rangle_{X_i} - \langle V, W \rangle_{X_i}^2}} - \frac{\partial_l \langle V, W \rangle_{X_i}}{\sqrt{\langle V, V \rangle_{X_i} \langle W, W \rangle_{X_i} - \langle V, W \rangle_{X_i}^2}}. \end{aligned} \quad (17)$$

Next, consider the following 10-by-10 matrix M :

$$\begin{pmatrix} -7.526 & 0.793 & -0.228 & -0.264 & -0.782 & 1.372 & 1.173 & -0.098 & 0.192 & 2.158 \\ 0.684 & 4.991 & -0.344 & -0.104 & -1.468 & -0.260 & 0.208 & -0.661 & -1.528 & 0 \\ -0.203 & -0.356 & 3.913 & -1.008 & -0.104 & 0 & -0.091 & -0.588 & -0.017 & 0.224 \\ -0.255 & -0.116 & -1.093 & 4.665 & -0.847 & -0.035 & 0.117 & -0.133 & -0.037 & 0 \\ -0.753 & -1.639 & -0.112 & -0.843 & 5.270 & -0.642 & 0.820 & 0.015 & 0 & 0.870 \\ 1.112 & -0.244 & 0 & -0.029 & -0.540 & -2.245 & 0.162 & 0 & -0.369 & 1.354 \\ 1.006 & 0.207 & -0.087 & 0.104 & 0.730 & 0.172 & -9.148 & 0 & 0 & 3.639 \\ -0.086 & -0.674 & -0.580 & -0.121 & 0.013 & 0 & 0 & 5.063 & -1.681 & 0 \\ 0.128 & -1.185 & -0.013 & -0.026 & 0 & -0.305 & 0 & -1.277 & 3.455 & 0 \\ 1.751 & 0 & 0.204 & 0 & 0.733 & 1.355 & 3.444 & 0 & 0 & -11.599 \end{pmatrix}$$

We state the following lemma, which we will prove in the next section.

Lemma 4.4. *For all $z \in B_{10^{-18}}(\hat{z})$ and $i, j, k \in \{1, \dots, 10\}$, we have*

$$\left| \frac{\partial^2 \Theta_i}{\partial z_j \partial z_k}(z) \right| \leq 10^{14}.$$

Corollary 4.5. *For all $z \in B_{10^{-18}}(\hat{z})$, we have*

$$d\Theta|_z = M + E,$$

where M is the matrix above and E is a matrix with $\|E\|_\infty < 0.002$.

Proof. Let $r = 10^{-18}$. One can use high precision floating-point arithmetic to compute $d\Theta|_{\hat{z}}$ and check that $d\Theta|_{\hat{z}} = M + E'$, where $\|E'\|_\infty < 0.001$. It then suffices to show that $d\Theta|_z = d\Theta|_{\hat{z}} + E''$, where $\|E''\|_\infty < 0.001$. To see this, notice that $\|z - \hat{z}\| < r$. It follows that

$$\left| \frac{\partial \Theta_i}{\partial x_j}(z) - \frac{\partial \Theta_i}{\partial x_j}(\hat{z}) \right| \leq 10 \cdot r \cdot \sup_{k \in [10], w \in B} \left| \frac{\partial^2 \Theta_i}{\partial z_j \partial z_k}(w) \right| < 10^{-3}.$$

This finishes the proof. $\square$

Proof of Proposition 4.3. We first claim that $\|M(V)\| > 1.5$ for all unit vectors V . To see this, one can numerically show that the smallest root of the characteristic polynomial of $M^t M$ is greater than 2.25, where M^t denotes the transpose of M . It then follows that the smallest singular value of M is greater than 1.5 (see Remark 4.6).

Next, given a 10-by-10 matrix E with $\|E\|_\infty < 0.002$, we must have

$$\|E\|_2 \leq \|E\|_F \leq 10^2 \|E\|_\infty < 0.2.$$

Here, $\|E\|_2 = \sup_{\|V\|=1} \|E(V)\|$ is the spectral norm of E and $\|E\|_F$ is the Frobenius norm. It follows that

$$\|d\Theta|_z(V)\| = \|M(V) + E(V)\| \geq \|M(V)\| - \|E(V)\| \geq 1.5 - 0.2 > 1.$$

This shows that $\|d\Theta|_z(V)\| > 2\lambda$.

To see that the angle θ between $d\Theta|_z(V)$ and $d\Theta|_w(V)$ is less than $\frac{\pi}{3}$ for all $z, w \in B_{10^{-18}}(\hat{z})$, we first observe that $d\Theta|_w(V) = d\Theta|_z(V) + E'(V)$ where $\|E'(V)\| < 0.004$. Then, elementary Euclidean geometry gives us

$$\sin \theta \leq \frac{\|E'(V)\|}{\|d\Theta|_z(V)\|} \leq \frac{0.4}{1.5} < \frac{\sqrt{3}}{2}.$$

This shows that $\theta < \frac{\pi}{3}$. $\square$

Remark 4.6. We explain how we find that the singular values of M are greater than 1.5. Since the entries of $M^t M$ are obtained from taking sums and products of entries of M , we may compute them precisely using floating-point arithmetic. This further implies that we may compute precisely the coefficients of the characteristic polynomial χ . We find that the characteristic polynomial has 10 distinct roots x_i by finding $a_i, b_i \in \mathbb{R}$ such that $a_i < b_i$ and $\chi(a_i)\chi(b_i) < 0$. The intermediate value theorem then guarantees the existence of the root $x_i \in (a_i, b_i)$ for each of the ten pairs (a_i, b_i) . We omit the tedious computation.

4.2 Bounds on Angles and Hyperbolic Distances

Before proving Lemma 4.4, we need to first establish some bounds on the angles and hyperbolic distances of the embedded surfaces in $\mathcal{B}$.

Lemma 4.7. *Consider the closed ball $B_r(0) \subset \mathbb{R}^3$ of radius $r \in (0, 1)$ centered at the origin. Then, for all $X \in B_r(0)$, $V \in \mathbb{R}^3$, we have*

$$\|V\|^2 \leq \|V\|_X^2 \leq \frac{\|V\|^2}{(1-r^2)^2}. \quad (18)$$

Proof. The first inequality follows from monotonicity of $1/(1+x^2)$ on $[0, r]$:

$$\langle V, V \rangle_X = \frac{\langle V, V \rangle}{a_X} + \frac{\langle V, X \rangle^2}{a_X^2} \geq \frac{\|V\|^2}{a_X} \geq \|V\|^2.$$

The second inequality follows from a routine application of the Cauchy-Schwarz inequality and monotonicity of $1/(1-x^2)^2$ on $[0, r]$:

$$\langle V, V \rangle_X \leq \frac{\|V\|^2}{a_X} + \frac{\|V\|^2 \|X\|^2}{a_X^2} = \frac{\|V\|^2}{a_X^2} \leq \frac{\|V\|^2}{(1-r^2)^2}.$$

This completes the proof. $\square$

Corollary 4.8. *Given $S \in \mathcal{B}$ and an edge e connecting X to Y , let $V = Y - X$. Then, we have $\|V\|_X \in [0.5, 13]$.*

Proof. Denote by $\hat{V} = \hat{Y} - \hat{X}$, where $\hat{V}$ is an edge connecting $\hat{X}$ to $\hat{Y}$ in $\hat{S}$ that corresponds to V . We compute numerically using Equation (5) that $\|\hat{V}\| \in [0.509, 1.561]$. Then, since $\hat{S} \subset B_{0.79}(0)$, we must have $S \subset B_{0.8}(0)$ for all $S \in \mathcal{B}$, so applying Lemma 4.7 with $r = 0.8$ gives us the desired bounds. $\square$

Proposition 4.9. *Given $S \in \mathcal{B}$, let e be an arbitrary edge in S connecting X to Y that corresponds to the edge $\hat{e}$ in $\hat{S}$ connecting $\hat{X}$ to $\hat{Y}$. Denote by $l(e)$ and $l(\hat{e})$ the hyperbolic length of edges e and $\hat{e}$ respectively. Then, we have*

$$|l(e) - l(\hat{e})| \leq 1.6 \times 10^{-17}.$$

Proof. From Corollary 4.8 we know that $S \subset B_{0.8}(0)$. Also, we must have $\|X - \hat{X}\| \leq 10^{-18}$ and $\|Y - \hat{Y}\| \leq 10^{-18}$. To find bounds on the hyperbolic distance $d_g(X, \hat{X})$ between X and $\hat{X}$, recall that geodesics in $\mathbb{K}^3$ are straight lines. We can therefore parametrize the geodesic from X to $\hat{X}$ as $\gamma(t) = (1-t)X + t\hat{X}$, where $t \in [0, 1]$. It follows that

$$d_g(X, \hat{X}) = l(\gamma(t)) = \int_0^1 \|\gamma'\|_{\gamma(t)} dt = \int_0^1 \|\hat{X} - X\|_{\gamma(t)} dt \leq 8\|\hat{X} - X\| \leq 8 \times 10^{-18}$$

where the second to the last inequality comes from Lemma 4.7 and $1/(1-0.8^2)^2 \leq 8$. We similarly get $d_g(Y, \hat{Y}) \leq 8 \times 10^{-18}$. It follows that

$$\begin{aligned} l(e) &= d_g(X, Y) \leq d_g(X, \hat{X}) + d_g(\hat{X}, \hat{Y}) + d_g(\hat{Y}, Y) \leq 1.6 \times 10^{-17} + l(\hat{e}) \\ l(\hat{e}) &= d_g(\hat{X}, \hat{Y}) \leq d_g(\hat{X}, X) + d_g(X, Y) + d_g(Y, \hat{Y}) \leq 1.6 \times 10^{-17} + l(e) \end{aligned}$$

Combining these two equations gives us the desired result. $\square$

Corollary 4.10. *With the same setup as above, we have $l(e) \in [0.6, 2.1]$.*

Proof. Using the coordinates of $\hat{S}$, we compute that $l(\hat{e}) \in [0.63, 2.08]$. We explain how we establish the bound. Given two distinct points $X, Y \in \mathbb{K}^3$, the quadratic polynomial $\|(1-t)X + tY\|^2 = 1$ has two distinct real roots $t_1 < 0$ and $t_2 > 1$. Let $A = (1-t_1)X + t_1Y$ and $B = (1-t_2)X + t_2Y$. The hyperbolic distance $d_g(X, Y)$ is given by

$$d_g(X, Y) = \frac{1}{2} \ln \left(\frac{\|X - B\| \|Y - A\|}{\|X - A\| \|Y - B\|} \right) = \frac{1}{2} \ln \left(\frac{t_2 - t_1 t_2}{t_1 - t_1 t_2} \right).$$

We may express $\|(1-t)X + tY\|^2 = 1$ as $at^2 + bt + c = 0$, where

$$a = \langle Y - X, Y - X \rangle; \quad b = 2\langle X, Y - X \rangle; \quad c = \langle X, X \rangle - 1.$$

Denote by $\Delta = b^2 - 4ac$ the discriminant of the quadratic $at^2 + bt + c = 0$. One can check that $\Delta > 0$ if $X \neq Y$. It follows that

$$d_g(X, Y) = \frac{1}{2} \ln \left(\frac{\Delta^{1/2} - b - 2c}{-\Delta^{1/2} - b - 2c} \right) = \ln(\Delta^{1/2} - b - 2c) - \frac{1}{2} \ln(4c^2 + 4ac + 4bc) \quad (19)$$

Since the square root function is monotonic increasing, it suffices to find floating-point numbers x_1, x_2 such that $x_1^2 \leq x \leq x_2^2$ to show $\sqrt{x} \in [x_1, x_2]$. This allows us to bound the value of $\Delta^{1/2}$ within 10^{-32} . It follows that $\Delta^{1/2} - b - 2c \in [1.93, 6.3]$. We also directly computed that $4c^2 + 4ac + 4bc \in [0.62, 1.99]$. It remains to compute $\ln(x)$ for $x \in [0.62, 6.3]$, which gives us $l(\hat{e}) \in [0.63, 2.08]$. We refer to §5.1 for how we obtained this bound. The corollary then follows from Proposition 4.9. $\square$

Proposition 4.11. *The map*

$$\psi(a, b, c) = \frac{\cosh a \cosh b - \cosh c}{\sinh a \sinh b} \quad (20)$$

is 70-Lipschitz on $[0.6, 2.1]^3$.

Proof. The partial derivatives of ψ are given by

$$\frac{\partial \psi}{\partial a} = \frac{\cosh a \cosh c - \cosh b}{\sinh^2 a \sinh b}; \quad \frac{\partial \psi}{\partial b} = \frac{\cosh b \cosh c - \cosh a}{\sinh^2 b \sinh a}; \quad \frac{\partial \psi}{\partial c} = -\frac{\sinh c}{\sinh a \sinh b}.$$

Since $\sinh, \cosh, \tanh$ are all positive and monotonic increasing on $[0.5, 2.1]$, we have $|\frac{\partial \psi}{\partial c}| \leq \frac{\sinh 2.1}{\sinh^2 0.5} \leq 15$ and

$$\left| \frac{\partial \psi}{\partial a} \right| \leq \left| \frac{\cosh c}{\tanh a \sinh a \sinh b} \right| + \left| \frac{1}{\sinh^2 a \tanh b} \right| \leq \frac{\cosh 2.1 + 1}{\tanh 0.5 \sinh^2 0.5} \leq 42. \quad (21)$$

See §5.1 for finding bounds on the values of $\sinh x, \cosh x, \tanh x$ where $x = 0.5, 2.1$. Similarly, $|\frac{\partial \psi}{\partial b}| \leq 42$. This implies

$$\frac{|\psi(y) - \psi(x)|}{\|y - x\|} \leq \sup_I \sqrt{\left(\frac{\partial \psi}{\partial a} \right)^2 + \left(\frac{\partial \psi}{\partial b} \right)^2 + \left(\frac{\partial \psi}{\partial c} \right)^2} \leq \sqrt{42^2 + 42^2 + 15^2} \leq 70$$

for all $x, y \in [0.6, 2.1]^3$. $\square$

Corollary 4.12. *For all $S \in \mathcal{B}$ and an angle θ of a triangle in S , we have $|\sin \theta| \geq 0.24$.*

Proof. Let $\hat{\theta}$ be the corresponding angle of θ in $\hat{S}$. We compute using Equation (7) and (8) that $\hat{\theta}$ satisfies $\cos \hat{\theta} \in [-0.008, 0.96]$. Denote by a, b, c the length of the edges of the triangle that contains θ , where c corresponds to the opposite edge of θ . Denote by $\hat{a}, \hat{b}, \hat{c}$ the edge lengths in $\hat{S}$. From Proposition 4.9, we know that $|a - \hat{a}| \leq 1.6 \times 10^{-17}$. Similarly, $|b - \hat{b}| \leq 1.6 \times 10^{-17}$ and $|c - \hat{c}| \leq 1.6 \times 10^{-17}$. It follows that $|\sqrt{a^2 + b^2 + c^2} - \sqrt{\hat{a}^2 + \hat{b}^2 + \hat{c}^2}| \leq 2.8 \times 10^{-17}$. Also, the hyperbolic law of cosine implies

$$\cos \theta = \frac{\cosh a \cosh b - \cosh c}{\sinh a \sinh b} = \psi(a, b, c).$$

From Proposition 4.11 we know that ψ is 70-Lipschitz on $[0.6, 2.1]^3$. Also, Proposition 4.10 gives us $a, b, c, \hat{a}, \hat{b}, \hat{c} \in [0.6, 2.1]$, so $|\cos \theta - \cos \hat{\theta}| \leq 2 \times 10^{-15}$. This gives us $\cos \theta \in [-0.01, 0.961]$. It remains a routine computation to check that $|\sin \theta| \geq 0.24$. $\square$

4.3 Bounds on the Second-Order Partial

Given $S \in \mathcal{B}$, for a triangle $(i, j, k) \in F$, recall our notation $V = X_j - X_i$, $W = X_k - X_i$, z_i the z -coordinate of X_i , and θ_{ijk} the angle at X_i . Throughout this section, we write

$$u = \langle V, W \rangle_{X_i}; \quad v = \|V\|_{X_i}; \quad w = \|W\|_{X_i}; \quad \theta = \theta_{ijk} = \arccos(u(vw)^{-1}).$$

We remind the readers of the bounds we obtained from §4.2:

- $\|X_i\| \leq 0.8$ ($S \subset B_{0.8}(0)$);
- $\|V\|, \|W\| \leq 1.6$ ($S \subset B_{0.8}(0)$);
- $v, w \in [0.5, 13]$ (Corollary 4.8);
- $|\sin \theta| \geq 0.24$ (Corollary 4.12);
- $|z_i| \leq 0.8$ ($S \subset B_{0.8}(0)$).

We remark that the Cauchy-Schwarz inequality further implies $u \leq 1.6^2$.

Lemma 4.13. *The following bounds on the first-order partials hold for all $l = i, j, k$:*

1. $|\partial_l u| \leq 10^3$;
2. $|\partial_l v|, |\partial_l w| \leq 10^3$;
3. $|\partial_l \theta| \leq 10^7$.

Proof. To see $|\partial_l u| \leq 10^3$, Equation (13) and (14) imply

$$|\partial_i u| \leq \frac{4 \cdot 0.8}{1 - 0.8^2} + \frac{8 \cdot 0.8 \cdot 1.6^2}{0.36^2} + \frac{4 \cdot 0.8 \cdot 1.6^4}{0.36^3} \leq 10^3; \quad |\partial_l u| \leq \frac{2 \cdot 0.8}{0.36} + \frac{0.8 \cdot 1.6^2}{0.36^2} \leq 10^3$$

for $l = j$ and k . Here, the denominator $1 - \langle X_i, X_i \rangle$ is bounded below by $1 - 0.8^2 = 0.36$.

To see $|\partial_l v| \leq 10^3$, we first observe that $(2v)^{-1} \leq (2 \cdot 0.5)^{-1} = 1$. Then, Equation (15) and (16) imply

$$|\partial_i v| \leq \frac{4 \cdot 0.8}{0.36} + \frac{8 \cdot 0.8 \cdot 1.6^2}{0.36^2} + \frac{4 \cdot 0.8 \cdot 1.6^4}{0.36^3} \leq 10^3; \quad |\partial_j v| \leq \frac{4 \cdot 0.8}{0.36} + \frac{2 \cdot 0.8 \cdot 1.6^2}{0.36^2} \leq 10^3.$$

Moreover, we have $\partial_k v = 0$. This shows $|\partial_l v| \leq 10^3$ for all l . The argument for $|\partial_l w| \leq 10^3$ is symmetric, so we will omit it.

Finally, to see $|\partial_l \theta| \leq 10^7$, Equation (17) implies

$$|\partial_l \theta| = \frac{\partial_l(vw) \cos \theta - \partial_l u}{vw \sin \theta} \leq \frac{2 \cdot 1.6 \cdot 10^3 + 10^3}{0.5^2 \cdot 0.34} \leq 3.2 \cdot 10^6 \leq 10^7.$$

This completes the proof. $\square$

Lemma 4.14. *The following bounds on the second-order partials hold for all $l, m \in \{i, j, k\}$:*

1. $|\partial_{lm} u| \leq 10^4$;
2. $|\partial_{lm} v|, |\partial_{lm} w| \leq 10^7$.

Proof. To ease notations we write

$$a = 1 - \|X_i\|^2; \quad t_1 = \langle X_i, V \rangle; \quad t_2 = \langle X_i, W \rangle; \quad t_3 = \langle V, W \rangle; \quad t_4 = \|V\|^2.$$

From Cauchy-Schwarz inequality, all the t_i 's satisfy $|t_i| \leq 1.6^2$, and $a > 1 - 0.8^2 = 0.36$.

We start by computing the second-order partials. Equation (13) and (14) gives us

$$\begin{aligned} \partial_{ii} u &= \frac{2}{a} + \frac{2(t_3 - t_1 - t_2 + 8z_i^2 - 4z_i z_j - 4z_i z_k + z_j z_k)}{a^2} \\ &\quad + \frac{4(t_1 t_2 + 2t_1 z_i z_k - 4t_1 z_i^2 + 2t_2 z_i z_j - 4t_2 z_i^2 + 2t_3 z_i^2)}{a^3} + \frac{24t_1 t_2 z_i^2}{a^4}; \\ \partial_{ij} u &= \frac{-1}{a} + \frac{t_2 - 4z_i^2 + 3z_i z_k}{a^2} + \frac{4t_2 z_i^2}{a^3}; \quad \partial_{jk} u = \frac{1}{a} + \frac{z_i^2}{a^2}; \end{aligned}$$

and $\partial_{jj} u = \partial_{kk} u = 0$. $\partial_{ik} u$ follows from symmetry.

For the second-order partials of v , we have

$$\begin{aligned} \partial_{ii} v &= \frac{1}{av} + \frac{t_4 - 2t_1 + 8z_i^2 - 8z_i z_j + z_j^2}{a^2 v} + \frac{2(2t_4 z_i^2 + 4t_1 z_i z_j - 8t_1 z_i^2 + t_1^2)}{a^3 v} + \frac{12t_1^2 z_i^2}{a^4 v} - \frac{(\partial_i v)^2}{v}; \\ \partial_{ij} v &= \frac{-1}{av} + \frac{3z_i z_j - 4z_i^2 + t_1}{a^2 v} + \frac{4t_1 z_i^2}{a^3 v} - \frac{\partial_i v \partial_j v}{v}; \quad \partial_{jj} v = \frac{-(\partial_j v)^2}{v} + \frac{1}{av} + \frac{z_i^2}{a^2 v}; \end{aligned}$$

and all other partials vanish.

Next, we find bounds on all these partials. The partials $|\partial_{lm} u|$ are bounded above by $|\partial_{ii} u|$, where

$$|\partial_{ii} u| \leq \frac{2}{0.36} + \frac{6 \cdot 1.6^2 + 34 \cdot 0.8^2}{0.36^2} + \frac{4 \cdot 1.6^4 + 56 \cdot 1.6^2 \cdot 0.8^2}{0.36^3} + \frac{24 \cdot 1.6^4 \cdot 0.8^2}{0.36^4} \leq 10^4.$$

The partials $|\partial_{lm}v|$ are bounded above by $\partial_{ii}v$, where

$$|\partial_{ii}v| \leq \frac{1}{0.36 \cdot 0.5} + \frac{3 \cdot 1.6^2 + 17 \cdot 0.8^2}{0.36^2 \cdot 0.5} + \frac{24 \cdot 1.6^2 \cdot 0.8^2 + 2 \cdot 1.6^4}{0.36^3 \cdot 0.5} + \frac{12 \cdot 1.6^4 \cdot 0.8^2}{0.36^4 \cdot 0.5} + \frac{10^6}{0.5} \leq 10^7.$$

The argument for $|\partial_{lm}w| \leq 10^7$ is symmetric, so we will omit it. $\square$

Proof of Lemma 4.4. From Equation (17), we compute $\partial_{lm}\theta$ as

$$\begin{aligned} \partial_{lm}\theta &= \frac{\partial_{lm}(vw) \cos \theta - \partial_l(vw) \sin \theta \partial_m \theta - \partial_{lm}u}{(v^2w^2 - u^2)^{1/2}} - \frac{(\partial_l(vw) \cos \theta - \partial_lu) \partial_l(v^2w^2 - u^2)}{2(v^2w^2 - u^2)^{3/2}} \\ &= \frac{\partial_{lm}(vw) \cos \theta - \partial_l(vw) \sin \theta \partial_m \theta - \partial_{lm}u}{vw \sin \theta} - \frac{(\partial_l(vw) \cos \theta - \partial_lu) \partial_l(v^2w^2 - u^2)}{2v^3w^3 \sin^3 \theta}. \end{aligned}$$

Then, from $|\sin \theta| \geq 0.24$, $u \leq 1.6^2$, and $v, w \in [0.5, 13]$ we know that $|\partial_{lm}\theta|$ is bounded above by

$$\frac{2 \cdot 10^6 + 2 \cdot 13 \cdot 10^7 + 2 \cdot 13 \cdot 10^{10} + 10^4}{0.5^2 \cdot 0.24} + \frac{(2 \cdot 13 \cdot 10^3 + 10^3) \cdot (4 \cdot 13 \cdot 10^3 + 2 \cdot 13^2 \cdot 10^3)}{2 \cdot 0.5^6 \cdot 0.24^3},$$

which is less than 10^{14} . This completes the proof. $\square$

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5 Appendix

5.1 Computing the Exponential Function

In this section, we explain how we obtain bounds on the values of the exponential function, a transcendental function, using floating-point arithmetic. Let $S_n(x)$ be the n -th partial sum of the Taylor expansion of e^x centered at 0. That is,

$$S_n(x) = \sum_{k=0}^n \frac{x^k}{k!}; \quad e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} = \lim_{n \rightarrow \infty} S_n(x).$$

Write $f(x) = e^x$. Given $a \in \mathbb{Z}_{>0}$, on the interval $[-a, a]$ we have

$$|f^{(n+1)}(x)| = |e^x| = e^x \leq e^a \leq 3^a.$$

It follows that for all $x \in [-a, a]$, we have

$$|S_n(x) - e^x| \leq \frac{a^{n+1}3^a}{(n+1)!}. \tag{22}$$

We are now ready to explain the computer calculations from Corollary 4.10. The task is to find an interval $[a, b]$ such that $a \leq \ln x \leq b$ for some given $x \in [0.6, 6.3]$. By monotonicity of the natural log function, it suffices to check that $e^a \leq x \leq e^b$. In particular, we noticed

that $e^{-2} \leq 2^{-2} = 0.25 \leq 0.6$ and $e^2 \geq 2.7^2 \geq 7.2 \geq 6.3$. This implies for all x such that $e^x \in [0.6, 6.3]$, we have $x \in [-2, 2]$. Then, using Equation (22), we computed up to 10^4 -digit of precision in `mpmath` that $|S_{20}(x) - e^x| \leq 10^{-10}$ for all $x \in [-2, 2]$. This justifies our calculation in Corollary 4.10.

In Proposition 4.11 we computed $\sinh x, \cosh x, \tanh x$ for $x = 0.5$ and $x = 2.1$. Using Equation (22), we see that $|S_{20}(x) - e^x| \leq 10^{-8}$ for $x \in [-3, 3]$. Denote by

$$S(x) = \frac{S_{20}(x) - S_{20}(-x)}{2}; \quad C(x) = \frac{S_{20}(x) + S_{20}(-x)}{2}; \quad T(x) = \frac{S_{20}(x) - S_{20}(-x)}{S_{20}(x) + S_{20}(-x)}.$$

It follows that $|S(x) - \sinh x| \leq 10^{-8}$ and $|C(x) - \cosh x| \leq 10^{-8}$ for all $x \in [-3, 3]$. Also, observe that $|\sinh x| \leq \cosh x \leq 3^3 = 27$ for $x \in [-3, 3]$, and $\cosh x \geq 1$ by AM-GM inequality, we have that

$$\begin{aligned} |T(x) - \tanh x| &= \left| \frac{S(x)}{C(x)} - \frac{\sinh x}{\cosh x} \right| \leq \frac{|\cosh x| |S(x) - \sinh x| + |\sinh x| |C(x) - \cosh x|}{|C(x)| |\cosh x|} \\ &\leq \frac{2 \cdot 27 \cdot 10^{-8}}{0.9} \leq 10^{-6}. \end{aligned}$$

We then computed $S(x)$, $C(x)$ and $T(x)$ for $x = 0.5$ and $x = 2.1$ to get

$$\begin{aligned} \sinh 0.5 &\in [0.521, 0.522] & \cosh 0.5 &\in [1.127, 1.128] & \tanh 0.5 &\in [0.462, 0.463] \\ \sinh 2.1 &\in [4.021, 4.022] & \cosh 2.1 &\in [4.144, 4.145] & \tanh 2.1 &\in [0.970, 0.971] \end{aligned}$$

5.2 The Coordinates for Robust Flatness

Here we provide the coordinates Y_{i,n_j} as promised in the proof of Proposition 3.2. We do so vertex-by-vertex.

For vertex 0, we have $d_0 = 9$ and $(n_0, \dots, n_8) = (1, 7, 8, 5, 3, 6, 4, 9, 2)$:

$Y_{0,1}$	69773419400785390350719255734910	0
$Y_{0,7}$	22673601164050404525981471083113	30522649530701644390945437817321
$Y_{0,8}$	-4502266020035690951698303492671	49787214971963367839502497819870
$Y_{0,5}$	-27890676491289921725823658542527	45152905884172442060397174220700
$Y_{0,3}$	-73673356961053011446423416051151	12554263676803953556985351161533
$Y_{0,6}$	-65716941293368369462892421315788	-20094064505362122337999561399511
$Y_{0,4}$	-63585776023578037908056362846628	-44491862575253747132630394705349
$Y_{0,9}$	-20681721190011172610982530346788	-48616891910058401869453205847887
$Y_{0,2}$	34828624619056461285801375727185	-44302110482568969803994230453983

For vertex 1, we have $d_1 = 8$ and $(n_0, \dots, n_7) = (0, 2, 4, 6, 5, 8, 3, 7)$:

$Y_{1,0}$	69773419400785390350719255734910	0
$Y_{1,2}$	60048146452422302324334406903101	34003898428632477407338918008929
$Y_{1,4}$	684253873310370060705932765911	45011277063511932691880974039935
$Y_{1,6}$	-38797441527227489121142967415251	35904267659650216159151157407406
$Y_{1,5}$	-34418586117854334955122669640154	-13661872610266486281752985853784
$Y_{1,8}$	-13188192204919717548387089195338	-32334046201585819117670137142096
$Y_{1,3}$	27532806238161586916092515324102	-59644150643141823546483370604325
$Y_{1,7}$	61207051193049841346494694029781	-20774055887376880717112213475032

For vertex 2, we have $d_2 = 8$ and $(n_0, \dots, n_7) = (0, 9, 6, 3, 8, 7, 4, 1)$:

$Y_{2,0}$	56353438990578855635490869491526	0
$Y_{2,9}$	48092387991317990941147748740637	39405069223657519928257610119072
$Y_{2,6}$	36264020972833155000907130948793	56108711187827723236898114831237
$Y_{2,3}$	-8023794381899623805048972165114	49766168048906812726617349210993
$Y_{2,8}$	-46688768526075731314833760652532	14579781429380556985176995990071
$Y_{2,7}$	-34323173836899021622431417499987	-16589459188636414317131134542419
$Y_{2,4}$	15801788573781581976615289172374	-66413096470080882136234194648455
$Y_{2,1}$	40340533719441891677134752026234	-55988269663804427444966060569426

For vertex 3, we have $d_3 = 8$ and $(n_0, \dots, n_7) = (0, 5, 4, 7, 1, 8, 2, 6)$:

$Y_{3,0}$	74735353497374034571180561502097	0
$Y_{3,5}$	64951355825035318887740051750196	27775730204660691498207963703796
$Y_{3,4}$	18203731888080621301584871054039	44282741941529823958117353691764
$Y_{3,7}$	-50270269905279614476683928736812	43757175563606495111050506045891
$Y_{3,1}$	-65679523751189539111583804355939	-1296258117380181104756639335080
$Y_{3,8}$	-53135278703958021717913038077077	-20673081744056204160401703682562
$Y_{3,2}$	-21947094098998016793187902172829	-45380368213190643026673636310429
$Y_{3,6}$	41407235240572972447410876062927	-39369780957805212274982571094628

For vertex 4, we have $d_4 = 8$ and $(n_0, \dots, n_7) = (0, 6, 1, 2, 7, 3, 5, 9)$:

$Y_{4,0}$	77605906656232659918496595614600	0
$Y_{4,6}$	35287460699942822202627864783646	31260480461897652641282558745742
$Y_{4,1}$	-12247171417163211623784470557145	43318472486111465943829115437047
$Y_{4,2}$	-66900524329443115864714765994097	13591017229050724080290741729896
$Y_{4,7}$	-71986590683672407001678934329849	-6598661163810768934712510249647
$Y_{4,3}$	-28248252874446950359042214693411	-38657124803256481078874090619402
$Y_{4,5}$	39577284522237084988918330394325	-48198246080382034917047601058065
$Y_{4,9}$	58114222516063173451633323816533	-23550433384261927623553718279566

For vertex 5, we have $d_5 = 7$ and $(n_0, \dots, n_6) = (0, 8, 1, 6, 9, 4, 3)$:

$Y_{5,0}$	53072353866459763807316258600612	0
$Y_{5,8}$	15132621870398054709768798392140	26932978119784418547635926675570
$Y_{5,1}$	-17271886847456450664911538801150	32756186565288176500115728267363
$Y_{5,6}$	-53366672941583648447918823670646	-264475383850852598262292837658
$Y_{5,9}$	-39121047935469870027816023468487	-22605253051769535765860271234150
$Y_{5,4}$	507268248785703193038086141638	-62363250831743394667763519936132
$Y_{5,3}$	30653902634212146206451278379057	-63643601918847202789375850542438

For vertex 6, we have $d_6 = 7$ and $(n_0, \dots, n_6) = (0, 3, 2, 9, 5, 1, 4)$:

$Y_{6,0}$	68720359438100137565590539767661	0
$Y_{6,3}$	25390557772908761618321852158345	51184551955783422998211990124352
$Y_{6,2}$	-6813962085088593529842703558335	66459285348099568242306767088465
$Y_{6,9}$	-29304204462392389437047998194621	15467804412244368520337457391781
$Y_{6,5}$	-49711247331825269118889114937918	-19412975475096919280111640378696
$Y_{6,1}$	-32118833053426954804131807462710	-41984979082054198551559535804433
$Y_{6,4}$	5307487555958890405906055277685	-46842855348516443461783670226668

For vertex 7, we have $d_7 = 6$ and $(n_0, \dots, n_5) = (0, 1, 3, 4, 2, 8)$:

$Y_{7,0}$	38022681706061569845515613988920	0
$Y_{7,1}$	11818625227981235011001127193086	63546712043039705538025314800530
$Y_{7,3}$	-37070064618057322788690894186616	55385925638802835554058530315113
$Y_{7,4}$	-60394084412598094355018887265961	39726139195463625691762018099811
$Y_{7,2}$	-35291354301407287983444968408251	-14415641851196358579951401594919
$Y_{7,8}$	6448073384057528034696596840928	-37852482326636002546581327669442

For vertex 8, we have $d_8 = 6$ and $(n_0, \dots, n_5) = (0, 7, 2, 3, 1, 5)$:

$Y_{8,0}$	49990370812584110646810728859197	0
$Y_{8,7}$	28931837076682989366588153660884	25245531723906183515103808022266
$Y_{8,2}$	5341182688010834270565922246268	48619778902245980212782746138780
$Y_{8,3}$	-36229090326845443500768801226589	44024847141506336954558235705473
$Y_{8,1}$	-26802550827068733629932236460907	-22383972537567782704648869515785
$Y_{8,5}$	7930519585113553825600252467951	-29857803237736129523910125242990

For vertex 9, we have $d_9 = 5$ and $(n_0, \dots, n_4) = (0, 4, 5, 6, 2)$:

$Y_{9,0}$	52833093515103383889224641548820	0
$Y_{9,4}$	940902316584597264618732735964	62697691138220619249446404872648
$Y_{9,5}$	-38467987934475596300054436654094	23699530826092470056977361403538
$Y_{9,6}$	-20279822767601932923431973551822	-26205307878821013050922465367016
$Y_{9,2}$	43525294101239515382128354128724	-44398040909811691945973843164387